Cyclistic Bike-Share Analysis
Understanding Member and Casual Rider Behaviour
Marketing Strategy Case Study - August 2025 – July 2026

Data source: Divvy public trip data
Google Data Analytics Capstone Case Study

Tools: Python · SQL (DuckDB) · Microsoft Excel

Prepared by:
Vinicius Loredam

2026

1. BUSINESS TASK

Identify how annual members and casual riders use *Cyclistic* bikes differently and use these insights to inform marketing strategies aimed at converting casual riders into annual members.

2. DATA & METHODOLOGY

Twelve months of ride records, covering the period from **August 2025** to **July 2026**, were analyzed. Each record represents a ride and contains information such as *ride_id*, bike type, start and end date and time, start and end stations, and customer type (member/casual).

The data were obtained from the official public Divvy trip data repository and were provided by Motivate International Inc. Twelve monthly CSV files were used, containing more than 6 million ride records. Due to the volume of data, **Python** and **SQL** (**DuckDB**) were used for data validation, preparation, aggregation, and analysis. Key descriptive statistics, including mean, median, and quartiles, were calculated during this stage. **Microsoft Excel** was subsequently used to review the results, perform supplementary calculations, and develop the visualizations.

During the validation phase, the monthly datasets were checked for structural consistency, including compatible columns and data types. Potential duplicates were also checked based on *ride_id*. Ride duration was calculated as the difference, in seconds, between the end and start timestamps (*ended_at* - *started_at*) and subsequently converted to minutes. Rides with negative durations were excluded from the analysis. The resulting ride classifications were validated for each of the 12 monthly datasets to ensure that all valid rides were accounted for in the analysis. The original data and files were preserved and were not modified.

The data were aggregated by customer type (member/casual), using additional grouping parameters such as month, bike type, and ride-duration intervals. Key descriptive statistics were calculated during this stage, along with absolute and relative counts and proportions.

The aggregated results were exported to Microsoft Excel, where the resulting tables were reviewed, analyzed, and used to develop the visualizations.

3. ANALYSIS & KEY FINDINGS

Finding #1 — Start and End Hotspots: Members vs. Casual Riders

A total of **2,033 unique station IDs** were identified during the analysis. The stations were ranked based on the total number of rides that started or ended at each station, and the 15 highest-volume stations for casual and member riders are presented below.

The 15 highest-volume stations represent substantial activity within the system, with 527,424 station events recorded for casual riders and 542,881 for members, indicating that these stations represent important locations within the system. However, only three stations are shared between the top 15 lists for casual and member riders: **Navy Pier**, **Wells St & Concord Ln**, and **Clark St & Armitage Ave** (see Table 1).

TOP 15 High Volume Stations					
Casual Riders			**Member Riders**		
Rank	**Station**	**Events**	**Rank**	**Station**	**Events**
1	**Navy Pier**	**110807**	1	Canal St & Madison St	**45286**
2	DuSable Lake Shore Dr & Monroe St	**62378**	2	State St & Chicago Ave	**43831**
3	Michigan Ave & Oak St	**45538**	**3**	**Wells St & Concord Ln**	**39480**
4	DuSable Lake Shore Dr & North Blvd	**41494**	4	Clinton St & Madison St	**39463**
5	Millennium Park	**37394**	5	Wells St & Elm St	**39060**
6	Theater on the Lake	**34041**	6	Clinton St & Jackson Blvd	**38570**
7	Shedd Aquarium	**32052**	7	Wells St & Huron St	**35867**
8	Dusable Harbor	**27328**	8	Clark St & Elm St	**35721**
9	Montrose Harbor	**20697**	9	Clinton St & Washington Blvd	**34543**
10	Michigan Ave & 8th St	**20552**	10	Wells St & Hubbard St	**33885**
11	Adler Planetarium	**19802**	11	University Ave & 57th St	**32468**
12	Clark St & Lincoln Ave	**19554**	12	Kingsbury St & Erie St	**31567**
13	**Wells St & Concord Ln**	**18947**	13	Desplaines St & Kinzie St	**31406**
14	**Clark St & Armitage Ave**	**18903**	**14**	**Navy Pier**	**31088**
15	Field Museum	**17937**	**15**	**Clark St & Armitage Ave**	**30646**
	Total Events:	**527424**		**Total Events:**	**542881**

Table 1: Highlighted stations appear in the Top 15 lists for both customer groups. Events represent rides recorded as either a station start or station end.

The composition of the top 15 also shows a notable difference between the two groups. Several of the highest-volume casual stations are located near major attractions and recreational areas, while member activity is more commonly concentrated at stations located at urban street intersections.

A substantial number of rides also contained unidentified station locations, with approximately one-third of rides having at least one missing station identifier. Recurring unidentified locations were observed throughout the analysis period, suggesting that the station

data contain additional spatial information that could potentially be recovered or better integrated.

Data interpretation note: These patterns do not establish that casual riders are tourists or that members primarily use the system for commuting. However, trip purpose cannot be determined from the available data, making these plausible hypotheses that would require additional data to evaluate.

Finding #2 — Seasonality & Customer Composition

The data show strong seasonal variation in ride volume, with the highest-volume month recording nearly six times as many rides as the lowest-volume month (see Figure 1). This seasonal effect impacts not only overall ride volume but also the composition of the customer groups. During the analyzed period, members accounted for approximately **64.4% of all rides**. The monthly share of member rides ranged from approximately **59% in July 2026 to 82% in January 2026**.

Casual riders accounted for a significantly larger share of rides during periods of high demand, while members represented a larger share of rides during periods of low demand. **Data interpretation note:** Because the dataset identifies individual rides rather than individual riders, these percentages represent the **share of rides**, not the proportion of individual *Cyclistic* users in each customer group.

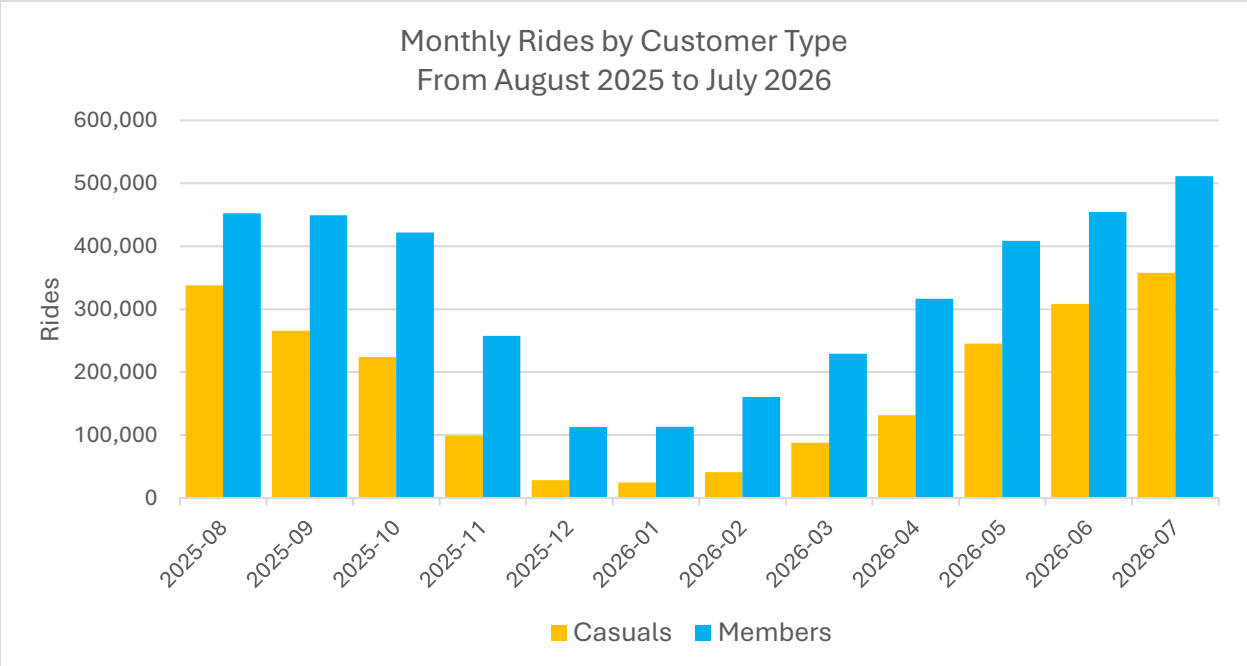


Figure 1: Monthly rides by customer type, from August 2025 to July 2026.

Finding #3 — Ride Duration & Distribution

There is a strong concentration of short-duration rides: **91.5% of rides lasted 30 minutes or less, while 98% lasted up to one hour**. Among the four combinations of customer type and bike type analyzed, **casual riders using classic bikes had the longest rides throughout the period**, with an overall mean duration of approximately **39 minutes**, more than **2.5 times the second-highest mean of approximately 15 minutes**. The median was also the highest, at approximately **16 minutes**, compared with less than 10 minutes for the other combinations.

Despite this, casual classic rides represented the **smallest share of total ride volume**, ranging from approximately **4% to 15% by month**, following the observed seasonal pattern.

Finding #4 — Electric Bikes

Electric bikes accounted for **68.7% of all rides**, representing more than **4 million rides** during the 12-month period. Their share was similar across customer groups, accounting for **71.5% of casual rides and 67.1% of member rides**.

Ride duration also differed between casual and member riders using electric bikes, although the gap was considerably smaller than among classic-bike rides. Casual electric rides averaged approximately **14 minutes**, compared with **11 minutes for members**, while median durations were **9.7 and 8.3 minutes**, respectively.

Electric-bike ridership also increased in both absolute volume and share between **August 2025 and July 2026**, although these months represent different points in the seasonal cycle.

4. RECOMMENDATIONS

Recommendation #1 — Target high-volume casual-use locations during the high-demand season

Evidence: Casual riders account for a substantially larger share of ride activity during the warmer, higher-demand period, while casual rides are more concentrated at a smaller number of high-volume stations, including several locations near major attractions and recreational areas.

Action: Begin membership-focused campaigns in late winter or early spring and prioritize high-volume casual-use locations as ridership enters the high season.

Rationale: Aligning campaign timing with increasing casual ridership and focusing outreach where casual activity is already concentrated may provide a more targeted approach to reaching prospective members.

The optimal campaign timing, locations, and messaging should be validated through marketing experiments, as the current dataset does not contain conversion or campaign-response data.

Recommendation #2 — Leverage the strong adoption of electric bikes

Evidence: Electric-bike rides accounted for the majority of rides in every month analyzed, with their monthly share remaining relatively stable at approximately 64.5%–72%, and remained the dominant bike type across both customer groups throughout the year.

Action: Consider incorporating electric-bike availability and value more prominently into membership marketing campaigns, given the high level of electric-bike usage among both casual riders and members.

Rationale: Because electric bikes account for the majority of observed ride activity, their availability and perceived value may represent an important component of the membership proposition.

Next step: Evaluate the operational feasibility and profitability of increasing electric-bike availability and charging capacity before making fleet or infrastructure investments.

Recommendation #3 — Strengthen user-level data and conversion measurement

Evidence: Current trip-level analysis suggests that ride duration could be explored as a segmentation variable, but user-level data would be required to determine whether longer-duration or more frequent casual riders represent a meaningful conversion opportunity.

Action: Collect or make available anonymized user-level usage, membership, and conversion data to enable customer segmentation and evaluate which behaviours are associated with membership adoption.

Rationale: This would allow *Cyclistic* to move from descriptive ride-level comparisons to identifying recurring casual riders, measuring conversion opportunities, and evaluating the effectiveness of specific marketing interventions.

5. LIMITATIONS & ADDITIONAL DATA

The primary limitation of this analysis is that the dataset identifies individual rides through *ride_id* but does not provide a persistent identifier for individual users. As a result, the analysis is conducted at the ride level rather than the customer level, limiting our ability to measure individual usage frequency, repeat behaviour, longitudinal usage patterns, or identify which casual riders may represent stronger membership-conversion opportunities.

The analysis of high-volume stations revealed distinct spatial patterns between member and casual riders. However, this analysis was also limited by the ride-level nature of the data and by the fact that approximately one-third of rides had at least one unidentified station location. Recurring unidentified locations were observed throughout the analysis period, suggesting that additional spatial information may be present but could not be reliably incorporated into the station analysis.

The 12-month analysis period provides a useful view of seasonal variation but limits the ability to evaluate recurring seasonal patterns and year-over-year trends. Observed changes may therefore reflect seasonal effects or period-specific factors that cannot be separated with the available timeframe.

The dataset also does not include information on ride or membership revenue, pricing, or operating costs. Consequently, the analysis cannot evaluate profitability by customer or bike type or determine the financial impact of specific membership strategies.

Additional anonymized user-level data, including individual ride frequency, membership history, and conversion outcomes, would allow *Cyclistic* to identify higher-value casual segments and measure the effectiveness of targeted marketing initiatives. Improved station identification would also support more complete spatial analysis and location-based marketing. Financial and operational data would further support the evaluation of the business impact and profitability of potential recommendations.

6. CONCLUSION

So, how do annual members and casual riders use *Cyclistic* bikes differently? Well, the analysis identified meaningful differences in where, when and how members and casual riders use the system, including distinct spatial patterns, strong seasonality, differences in ride duration, and a high adoption of electric bikes among both groups.

However, these insights do not tell us which casual riders are more likely to become annual members, or which marketing interventions are most effective at driving conversion. The available data do not provide user-level information or conversion history, limiting our ability to

move from aggregate behavioural patterns to individual customer targeting and measurable conversion strategies.

Additional data, such as user-level information, more complete and reliable station identification, and cost and revenue data would provide a better understanding of rider behaviour and a stronger basis for measuring the effectiveness of marketing campaigns.